
Subject: Dissertation Proposal: Multimodal RAG for Explainable Breast Cancer Diagnosis
(Tharit Khan)

Dear [Supervisor Name],

I hope you are having a good week.

I am writing to propose a dissertation topic for my MSc in Artificial Intelligence. Drawing on my recent industry experience building RAG pipelines at KPIT and my current work with PyTorch vision models, I would like to develop a **Multimodal Retrieval-Augmented Generation (RAG)** system for oncology.

The core objective is to move beyond "Black Box" classification. Instead of simply predicting cancer probability, I aim to build a system that retrieves historically similar patient cases—matched by both **MRI features** and **molecular biomarkers**—to generate explainable, evidence-based diagnostic reports.

Below is the outline of the proposed project, spanning the required 3-month (600-hour) timeline.

Proposal Title

"Enhancing Diagnostic Trust: A Multimodal RAG Framework Integrating MRI Imaging and Genomic Biomarkers for Breast Cancer Diagnosis."

1. The Problem & Solution

- **The Problem:** Standard Deep Learning models (like ResNet) offer high accuracy in detecting tumors but lack interpretability. They cannot explain *why* a decision was made, nor can they integrate critical non-visual data like blood/tissue markers.
- **The Proposed Solution:** A "Hybrid Retrieval" system.
 1. **Input:** A patient's MRI scan + Tabular Biomarkers (ER, PR, HER2, Ki-67).
 2. **Retrieval:** The system queries a vector database to find the top- k historical patients who are **visually similar** (via Image Embeddings) AND **biologically matched** (via Metadata Filtering).
 3. **Generation:** An LLM (GPT-4o/Claude) synthesizes a diagnostic report, citing the outcomes of these retrieved "Digital Twins" to justify the prognosis.

2. Dataset

I have selected the **Duke Breast Cancer MRI Dataset** (publicly available via TCIA).

- **Size:** 922 Patients.
- **Why this dataset?** It is uniquely suited for this multimodal approach as it provides

matched pairs of:

- **Imaging:** Pre-operative Dynamic Contrast-Enhanced (DCE) MRI.
- **Clinical Data:** Verified status for **Estrogen Receptor (ER)**, **Progesterone Receptor (PR)**, **HER2**, and **Ki-67** (proliferation index).

3. Methodology & Tech Stack

- **Visual Encoder:** Fine-tune a **ResNet50** or **VGG19** (using PyTorch) to extract dense vector embeddings from the MRI slices, stripping the final classification layer.
- **Vector Database:** Use **FAISS** or **Qdrant** to store these image vectors. Crucially, I will implement **Hybrid Search** logic to filter results based on the tabular biomarker data (e.g., *"Find visually similar tumors that are also HER2-Positive"*).
- **Generator Agent:** Use **LangChain** to feed the clinical notes of the retrieved neighbors into an LLM, generating a patient-friendly explanation of the diagnosis (e.g., explaining how High Ki-67 levels influence the treatment path).

4. Project Timeline (3 Months / 600 Hours)

- **Month 1: Infrastructure & Embedding**
 - Data ingestion and cleaning (matching MRI NIfTI files to Clinical CSVs).
 - Training the ResNet backbone on the Duke dataset to ensure high-quality feature extraction.
 - Building the "Index" of 922 patient vectors.
- **Month 2: The RAG Pipeline**
 - Implementing the "Hybrid Search" algorithm (Vector Similarity + Metadata Filters).
 - Developing the LLM prompting strategy ("In-Context Learning" using retrieved case histories).
 - Prototyping a simple UI (Streamlit) to visualize the "Top-5 Retrieved Cases" side-by-side.
- **Month 3: Evaluation & Thesis Writing**
 - **Quantitative Eval:** Measuring **Precision@K** (Do the retrieved neighbors actually share the same diagnosis/survival outcome?).
 - **Qualitative Eval:** Assessing the hallucination rate and clarity of the generated reports.
 - Final thesis writing and documentation.

Feasibility

I believe this is highly feasible given my background. I have already built RAG systems for test-script generation at KPIT and am currently fine-tuning VGG19 models for CT scans in my personal projects. This dissertation effectively combines those two existing skill sets into a novel medical application.

I would value your feedback on this scope. Do you think the focus on the **Duke Dataset** is

appropriate, or would you suggest narrowing the scope further?

Best regards,

Tharit Khan

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